

Derivation of the Duan (1995) GARCH and NGARCH Model Under the Risk-Neutral Measure

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1 Physical-measure GARCH model

Let the asset price at time t be S_t , and define the log-return

$$r_{t+1} = \log S_{t+1} - \log S_t.$$

Under the physical probability measure $\mathbb{P}$, returns follow a GARCH(1,1) process:

$$r_{t+1} = \mu + \sqrt{h_{t+1}} z_{t+1}, \quad z_{t+1} \sim \mathcal{N}(0, 1),$$

with conditional variance dynamics

$$h_{t+1} = \omega + \alpha(r_t - \mu)^2 + \beta h_t,$$

where $\omega > 0$, $\alpha \geq 0$, $\beta \geq 0$, and $\alpha + \beta < 1$.

2 No-arbitrage condition

In discrete time, absence of arbitrage implies the existence of a probability measure $\mathbb{Q}$ such that the discounted asset price is a martingale:

$$\mathbb{E}^{\mathbb{Q}}[S_{t+1} \mid \mathcal{F}_t] = e^{r_f} S_t,$$

where r_f is the continuously compounded risk-free rate over one period.

Using $S_{t+1} = S_t e^{r_{t+1}}$, this condition is equivalent to

$$\mathbb{E}^{\mathbb{Q}}[e^{r_{t+1}} \mid \mathcal{F}_t] = e^{r_f}.$$

3 Duan's Local Risk-Neutral Valuation Relationship

Duan (1995) introduces the *Local Risk-Neutral Valuation Relationship (LRNVR)*, which is based on the following assumptions:

1. The conditional variance process is invariant under the change of measure:

$$h_{t+1}^{\mathbb{Q}} = h_{t+1}^{\mathbb{P}}.$$

2. The standardized innovations remain Gaussian:

$$z_{t+1}^{\mathbb{Q}} \sim \mathcal{N}(0, 1).$$

3. Only the conditional mean of returns is adjusted to enforce the martingale condition.

4 Generic risk-neutral return specification

Assume that under $\mathbb{Q}$, returns have the form

$$r_{t+1} = \mu_t^{\mathbb{Q}} + \sqrt{h_{t+1}} z_{t+1}^{\mathbb{Q}}.$$

The objective is to determine $\mu_t^{\mathbb{Q}}$.

5 Imposing the martingale condition

Compute the conditional expectation under $\mathbb{Q}$:

$$\begin{aligned} \mathbb{E}^{\mathbb{Q}}[e^{r_{t+1}} \mid \mathcal{F}_t] &= \mathbb{E}^{\mathbb{Q}} \left[\exp \left(\mu_t^{\mathbb{Q}} + \sqrt{h_{t+1}} z_{t+1}^{\mathbb{Q}} \right) \mid \mathcal{F}_t \right] \\ &= \exp \left(\mu_t^{\mathbb{Q}} \right) \mathbb{E} \left[\exp \left(\sqrt{h_{t+1}} Z \right) \right], \end{aligned}$$

where $Z \sim \mathcal{N}(0, 1)$.

Using the moment-generating function of the standard normal distribution,

$$\mathbb{E}[e^{aZ}] = \exp \left(\frac{1}{2} a^2 \right),$$

we obtain

$$\mathbb{E}^{\mathbb{Q}}[e^{r_{t+1}} \mid \mathcal{F}_t] = \exp \left(\mu_t^{\mathbb{Q}} + \frac{1}{2} h_{t+1} \right).$$

6 Risk-neutral drift

Enforcing the martingale condition

$$\exp \left(\mu_t^{\mathbb{Q}} + \frac{1}{2} h_{t+1} \right) = e^{r_f},$$

yields

$$\mu_t^{\mathbb{Q}} = r_f - \frac{1}{2} h_{t+1}.$$

7 Risk-neutral GARCH dynamics

Substituting the risk-neutral drift into the return equation gives

$$\begin{aligned} r_{t+1} &= r_f - \frac{1}{2} h_{t+1} + \sqrt{h_{t+1}} z_{t+1}^{\mathbb{Q}}, \\ h_{t+1} &= \omega + \alpha(r_t - \mu)^2 + \beta h_t, \\ z_{t+1}^{\mathbb{Q}} &\sim \mathcal{N}(0, 1). \end{aligned}$$

This system defines the *Duan (1995) GARCH option pricing model*.

8 Option pricing

For a European-style payoff $\Phi(S_T)$ maturing at time T , the no-arbitrage price is

$$V_0 = e^{-r_f T} \mathbb{E}^{\mathbb{Q}}[\Phi(S_T)].$$

Since the distribution of S_T is not available in closed form, the expectation is evaluated using Monte Carlo simulation under the risk-neutral dynamics.

9 Physical Measure ($\mathbb{P}$) Dynamics in NGARCH

Duan's paper also starts with the non-linear GARCH (NGARCH) process, where the log-return is conditionally log-normal.

9.1 Asset Return Process

The log-return of the asset S_t is defined as:

$$\ln\left(\frac{S_t}{S_{t-1}}\right) = r_f + \lambda h_t - \frac{1}{2}h_t + \sqrt{h_t}z_t$$

where:

- r_f is the constant risk-free rate.
- λh_t is the equity risk premium.
- h_t is the conditional variance.
- z_t is the standardized innovation: $z_t \mid \mathcal{F}_{t-1} \sim N(0, 1)$ under $\mathbb{P}$.

9.2 Conditional Variance Process (NGARCH(1, 1))

The variance follows the non-linear GARCH structure, capturing the leverage effect:

$$h_t = \omega + \beta h_{t-1} + \alpha(z_{t-1} - \gamma\sqrt{h_{t-1}})^2$$

10 The LRNVR Transformation

The LRNVR seeks an equivalent martingale measure ($\mathbb{Q}$) by satisfying the no-arbitrage condition and maintaining the conditional distribution type.

10.1 Martingale Condition (Risk-Neutral Drift)

The expected gross return under $\mathbb{Q}$ must equal the gross risk-free rate:

$$E_{\mathbb{Q}}\left[\frac{S_t}{S_{t-1}} \mid \mathcal{F}_{t-1}\right] = e^{r_f}$$

Since $\ln(S_t/S_{t-1}) \sim N(\mu_t^{\mathbb{Q}}, h_t)$, we require $E_{\mathbb{Q}}[e^{\ln(S_t/S_{t-1})}] = e^{\mu_t^{\mathbb{Q}} + \frac{1}{2}h_t}$. Setting the exponents equal:

$$\mu_t^{\mathbb{Q}} + \frac{1}{2}h_t = r_f$$

The **Risk-Neutral Drift** is:

$$\mu_t^{\mathbb{Q}} = r_f - \frac{1}{2}h_t$$

10.2 Transformation of Innovations

To change the drift from $\mu_t^{\mathbb{P}}$ to $\mu_t^{\mathbb{Q}}$, the innovations are transformed via:

$$z_t = z_t^* - \lambda\sqrt{h_t}$$

where $z_t^* \mid \mathcal{F}_{t-1} \sim N(0, 1)$ under $\mathbb{Q}$.

11 Risk-Neutral Measure ($\mathbb{Q}$) Dynamics

11.1 Risk-Neutral Asset Return Process

Substitute z_t into the log-return equation:

$$\begin{aligned}\ln\left(\frac{S_t}{S_{t-1}}\right) &= r_f + \lambda h_t - \frac{1}{2}h_t + \sqrt{h_t}(z_t^* - \lambda\sqrt{h_t}) \\ \ln\left(\frac{S_t}{S_{t-1}}\right) &= r_f + \lambda h_t - \frac{1}{2}h_t + \sqrt{h_t}z_t^* - \lambda h_t\end{aligned}$$

The λh_t terms cancel:

$$\ln\left(\frac{S_t}{S_{t-1}}\right) = \left(r_f - \frac{1}{2}h_t\right) + \sqrt{h_t}z_t^* \quad (\text{Under } \mathbb{Q})$$

11.2 Risk-Neutral Conditional Variance Process

Substitute z_{t-1} into the NGARCH variance equation:

$$\begin{aligned}h_t &= \omega + \beta h_{t-1} + \alpha(z_{t-1} - \gamma\sqrt{h_{t-1}})^2 \\ h_t &= \omega + \beta h_{t-1} + \alpha((z_{t-1}^* - \lambda\sqrt{h_{t-1}}) - \gamma\sqrt{h_{t-1}})^2\end{aligned}$$

Grouping the $\sqrt{h_{t-1}}$ terms:

$$h_t = \omega + \beta h_{t-1} + \alpha(z_{t-1}^* - (\lambda + \gamma)\sqrt{h_{t-1}})^2 \quad (\text{Under } \mathbb{Q})$$

The risk premium λ is absorbed into the leverage term γ , creating $\gamma^* = \lambda + \gamma$.

12 Reference

Duan, J.-C. (1995). *The GARCH Option Pricing Model*. Mathematical Finance, 5(1), 13–32.